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ELECTROCHEMICAL APPARATUS, ELECTRONIC DEVICE, AND MANUFACTURING METHOD FOR ELECTROCHEMICAL APPARATUS

Abstract

The electrochemical apparatus includes a battery cell, a packaging bag, and a sealing piece. The battery cell is disposed in the packaging bag, and a tab of the battery cell protrudes from a sealing edge of the packaging bag. The sealing piece is disposed on both side of the tab, a top edge of the sealing piece is located outside the packaging bag, and a bottom edge is located inside the packaging bag. The top edge includes a first section, and a second section and a third section located at two ends of the first section. A length of the first section is A, a length of the bottom edge is B, and a width of the tab is C, meeting B>A≥C. A height of the second section or the third section is H, meeting 0.05 mm≤H≤0.5 mm.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of International Application No. PCT/CN2022/132454, filed on Nov. 17, 2022, the contents of which are incorporated herein by reference in its entirety.

TECHNICAL FIELD
[0002] This application relates to the field of battery technology, and in particular, to an electrochemical apparatus, an electronic device, and a manufacturing method.
BACKGROUND
[0003] As shown in FIGS. **14** to **16**, in the electrochemical apparatus of the prior art, the sealing piece is a conventional rectangle to ensure effective sealing between a tab and a packaging bag. However, during the hot pressing and sealing process of the conventional rectangular sealing piece, due to the double effects of high temperature softening and compression, part of the sealing piece exposed by the packaging bag tends to deform outward, causing two sides of a top end of the sealing piece to warp up, thus increasing the exposed length of the sealing piece. After the tab is bent, the warping part of the top end of the sealing piece causes the electrochemical apparatus to be subjected to interference between a battery and a battery compartment during the assembly of an electronic device. Alternatively, to prevent the warping part from being exposed, the bending position of the tab needs to be set back, increasing the overall length of the electrochemical apparatus and affecting the energy density. Additionally, the warping on two sides of the top end of the sealing piece is also likely to cause the sealing piece to protrude relative to the battery in the thickness direction of the battery, affecting the energy density.
SUMMARY
[0004] This application provides an electrochemical apparatus, an electronic device, and a manufacturing method to resolve the above technical problems.
[0005] An embodiment of this application provides an electrochemical apparatus, including a battery cell, a packaging bag, and a sealing piece. The battery cell includes an electrode plate and a tab connected to the electrode plate. The battery cell is disposed in the packaging bag, and the tab protrudes from a sealing edge of the packaging bag. The sealing piece is disposed on both side of the tab, and the packaging bag covers part of the sealing piece. The sealing piece includes a top edge and a bottom edge opposite each other, the top edge is located outside the packaging bag, and the bottom edge is located inside the packaging bag. The top edge includes a first section, a second section, and a third section, the second section and the third section are respectively located on two sides of the first section, and along a width direction of the tab, a length of the first section is A, a length of the bottom edge is B, and a width of the tab is C, meeting $B>A\geq C$. Along an extension direction of the tab, a distance from any point on the second section and the third section to the bottom edge is less than a distance between the first section and the bottom edge. Along the extension direction of the tab, a distance between two ends of the second section or the third section is a height H of the second section or the third section, where $0.05\ mm\leq H\leq 0.5\ mm$.
[0006] Thus, along the extension direction of the tab, the second section and the third section do not exceed the first section. The height difference between the second section and the first section

and the height difference between the third section and the first section can reduce the impact caused by the outward-extending deformation of the top edge. By limiting the ranges of the height difference between the second section and the first section and the height difference between the third section and the first section, the problem of the sealing piece warping in the thickness direction of the battery cell can be alleviated, thereby lowering the probability of two sides of the top end of the sealing piece extending outward and warping. This helps reduce the length and thickness of the electrochemical apparatus, thereby increasing the energy density. Additionally, by limiting the height ranges of the second section and the third section, the risk of electrolyte leakage caused by the second section and the third section entering the packaging bag can be reduced, thereby increasing the energy density of the electrochemical apparatus in a case of small electrolyte leakage rate losses of the electrochemical apparatus and alleviating the interference problem during assembly.

[0007] In a possible embodiment, 0 mm≤B−A≤8 mm. Through this arrangement, the length A of the first section is less than the length B of the bottom edge, and the second section and the third section can be formed on two sides of the first section. If the width of the bottom edge is constant, when the value of B−A is too large, the difference between the length of the first section and the width of the tab is too small, which is likely to cause cracking at the joint between the sealing piece and the tab during assembly, form a leakage channel, and increase the electrolyte leakage rate of the electrochemical apparatus.

[0008] In a possible embodiment, 1.8 mm≤B−A≤8 mm. Through this arrangement, the two ends of the first section can form distinct second section and third section, reducing the exposed size and warping amount of the sealing piece, alleviating interference problems during assembly, and reducing the loss of energy density of the electrochemical apparatus, while not causing a high electrolyte leakage rate of the electrochemical apparatus.

[0009] In a possible embodiment, 3 mm≤B−A≤7 mm. This arrangement helps further reduce the exposed size of the sealing piece without significantly damaging the sealing performance of the electrochemical apparatus.

[0010] In a possible embodiment, 0.15 mm≤H≤0.3 mm. This parameter range is also the height parameter range of the second section and/or the third section of the sealing piece after hot melting. This can further reduce the exposed size of the sealing piece, improve the energy density performance, alleviate leakage problems of the electrochemical apparatus, and meet subsequent assembly requirements.

[0011] In a possible embodiment, a distance from an end of the second section or the third section connected to the first section to the tab is W, where 0.3 mm≤W≤1.1 mm. This parameter range is also the parameter range of the shoulder width of the top edge of the sealing piece after hot melting of the sealing piece, which helps maintain the covering effect of the sealing piece on the tab and ensures the sealing performance between the tab and the packaging bag.

[0012] In a possible embodiment, a distance between a side edge of the sealing piece and a corresponding side edge of the tab is X, where 1.5 mm≤X≤4.5 mm. Through such arrangement, an end of the bottom edge of the sealing piece exceeds an end of the first section, increasing the contact area of the sealing piece in the packaging bag, which helps ensure the sealing effect between the tab and the packaging bag.

[0013] In a possible embodiment, along the extension direction of the tab, a distance between the first section and the bottom edge is a height Y of the sealing piece, where 3.5 mm≤Y≤12.5 mm. A suitable value may be chosen based on the size of the tab to maintain the sealing effect between the tab and the packaging bag while reasonably planning the height dimensions of the sealing piece, alleviating the problem of the increase in the overall size of the electrochemical apparatus caused by excessive exposure of the sealing piece.

[0014] In a possible embodiment, the second section is obliquely connected between an end of the first section and a side edge of the sealing piece, and the third section is obliquely connected

between another end of the first section and another side edge of the sealing piece. Thus, the second section and the third section form chamfered structures on two sides of the top edge of the sealing piece, and these simple structures can reduce the impact caused by the outward-extending deformation of the sealing piece.
[0015] In a possible embodiment, the second section is an arc section connected between an end of the first section and a side edge of the sealing piece, and the second section is recessed toward a side edge of the tab or the second section is bent toward an outer side of the sealing piece.
[0016] In a possible embodiment, the third section is an arc section connected between another end of the first section and another side edge of the sealing piece, and the third section is recessed toward the side edge of the tab or the third section protrudes toward the outer side of the sealing piece.
[0017] Thus, the end of the top edge of the sealing piece forms a rounded corner structure, and the second section and the third section do not exceed the first section along the extension direction of the tab. The rounded corner structure can reduce sharp parts at the end of the sealing piece, effectively reducing the warping of the sealing piece.
[0018] In a possible embodiment, the second section and/or the third section is provided with a depression, such that the second section and/or the third section includes a first subsection and a second subsection. The first subsection is located between the first section and the second subsection, the second subsection includes at least one section, an endpoint connecting the first subsection and the first section is M, an endpoint connecting the second subsection and the side edge of the sealing piece is N, the connection endpoint M and the connection endpoint N form a line segment MN, and an included angle between the first section and the line segment MN is greater than 90°. The included angle between the first section and the line segment MN being less than 90° causes the second section to warp. Setting the included angle between the first section and the line segment MN to be greater than 90° can reduce or even avoid the loss of energy density of the electrochemical apparatus.
[0019] In a possible embodiment, the second subsection includes at least two sections, and the included angle between adjacent sections is greater than or equal to 90°. The included angle being less than 90° can cause the second subsection to warp. Setting the included angle to be greater than 90° can reduce or even avoid the loss of energy density of the electrochemical apparatus.
[0020] In a possible embodiment, the second subsection includes n sections, the n-th section is connected to the side edge of the sealing piece, the distances from the intersections between the (n−1)th section close to the first subsection and the n-th section to the side edge of the tab do not exceed 2.5 mm. This multi-section arrangement allows the top edge of the sealing piece to fit better, improving the sealing performance of the electrochemical apparatus.
[0021] In a possible embodiment, a groove is further opened between the first subsection and the second subsection to absorb deformation of the sealing piece.
[0022] In a possible embodiment, along the extension direction of the tab, a distance between an end of the second section away from the first section and the sealing edge of the packaging bag and a distance between an end of the third section away from the first section and the sealing edge of the packaging bag are both greater than or equal to 0.1 mm. This arrangement can prevent the second section or the third section from entering the sealing region and affecting the sealing performance of the electrochemical apparatus.
[0023] In a possible embodiment, the electrochemical apparatus further includes a circuit board, where the circuit board is located on a side of the sealing piece and covers part of the sealing piece exposed by the packaging bag, and an end of the tab protruding from the packaging bag is bent toward the circuit board and connected to the circuit board.
[0024] In a possible embodiment, the packaging bag is rectangular, and along the extension direction of the tab, vertical distances from any points on the sealing edge to the first section of the top edge are identical. Thus, the shape of the sealing edge may be regular and stable, helping

increase the sealing effect.
[0025] An embodiment of this application provides an electronic device, which includes a housing, an electronic component connection line, and the electrochemical apparatus of the above embodiments. The electronic component connection line and the electrochemical apparatus are disposed in the housing, and the electronic component connection line is electrically connected to the electrochemical apparatus.
[0026] In a possible embodiment, the housing includes an accommodating cavity, the electrochemical apparatus is mounted in the accommodating cavity, and the top edge of the sealing piece of the electrochemical apparatus does not exceed the accommodating cavity.
[0027] An embodiment of this application provides a manufacturing method for electrochemical apparatus, used for manufacturing the above electrochemical apparatus. The manufacturing method for electrochemical apparatus includes: die-cutting the top edge of the sealing piece, such that the top edge of the sealing piece forms the first section, the second section, and the third section, where the second section and the third section are respectively located on two sides of the first section, and along the width direction of the tab, the length of the first section is A, the length of the bottom edge is B, and the width of the tab is C, meeting $B>A\geq C$; and along the extension direction of the tab, the distance from any point on the second section and the third section to the bottom edge is less than a distance between the first section and the bottom edge; [0028] welding the tab to the electrode plate, and winding or stacking the electrode plate to form the battery cell; [0029] preparing the tab, and adhering the sealing piece to the tab; [0030] placing the battery cell into the packaging bag, where the tab protrudes from the sealing edge of the packaging bag, the packaging bag covers part of the sealing piece, the top edge of the sealing piece is located outside the packaging bag, and the bottom edge is located inside the packaging bag; and [0031] hot pressing and sealing the packaging bag, where the sealing piece is sealingly connected between the packaging bag and the tab.
[0032] In a possible embodiment, in the step of hot pressing and sealing the packaging bag, a hot pressing pressure is 120±50 Kgf, and a hot pressing temperature is 190±5° C.
[0033] In a possible embodiment, a thickness of the sealing piece is 30 μm-80 μm.
[0034] In a possible embodiment, after the die-cutting the top edge of the sealing piece, the distance between two ends of the second section or the third section is a height H_1 of the second section or the third section, where $0.15\text{ mm}\leq H_1\leq 0.7\text{ mm}$; and after the step of hot pressing and sealing the packaging bag, the height of the second section or the third section is H, where $0.05\text{ mm}\leq H\leq 0.5\text{ mm}$. This arrangement can pre-reserve the extension position of the sealing piece during hot pressing, preventing the second section or the third section of the sealing piece from exceeding the first section of the sealing piece after the step of hot pressing and sealing the packaging bag. This helps reduce the warping of the sealing piece in the length direction of the electrochemical apparatus and the warping in the thickness direction, thereby reducing the loss of energy density.
[0035] In a possible embodiment, $0.3\text{ mm}\leq H_1\leq 0.5\text{ mm}$, and after the step of hot pressing and sealing the packaging bag, $0.15\text{ mm}\leq H\leq 0.3\text{ mm}$. This arrangement helps further reduce the warping amount of the sealing piece and increase the energy density of the electrochemical apparatus, and does not significantly compromise the leakage rate of the electrochemical apparatus.
[0036] In a possible embodiment, after the die-cutting of the top edge of the sealing piece, the difference between the length B_1 of the bottom edge of the sealing piece and the length A_1 of the first section is $0.2<B_1-A_1\leq 8.3$ mm. After the step of hot pressing and sealing the packaging bag, $0<B-A\leq 8$ mm. When $0.2<B_1-A_1\leq 8.3$, the extension position of the sealing piece can be pre-reserved during hot pressing, preventing the length of the first section of the sealing piece from exceeding the length of the bottom edge of the sealing piece after the step of hot pressing and sealing the packaging bag. This helps reduce the warping of the sealing piece in the length direction of the electrochemical apparatus and the warping in the thickness

direction, thereby reducing the loss of energy density.
[0037] In a possible embodiment, 2.1 mm≤B_1−A_1≤8.3 mm, and after the step of hot pressing and sealing the packaging bag, 1.8 mm≤B−A≤8 mm. This arrangement helps further reduce the warping amount of the sealing piece.
[0038] In a possible embodiment, 3.3 mm≤B_1−A_1≤7.3 mm; and after the step of hot pressing and sealing the packaging bag, 3 mm≤B−A≤7 mm. This arrangement helps further reduce the warping amount of the sealing piece and allows for a high energy density of the electrochemical apparatus, and does not significantly compromise the leakage rate of the electrochemical apparatus.
[0039] In a possible embodiment, after the die-cutting the top edge of the sealing piece, the length from the end of the second section or the third section connected to the first section to the tab is W, where 0.2 mm≤W**1**≤1.3 mm. After the step of hot pressing and sealing the packaging bag, 0.3 mm≤W≤1.1 mm.
[0040] In a possible embodiment, a distance between a side edge of the sealing piece and a corresponding side edge of the tab is X, where 0.3 mm≤X≤4.5 mm.
[0041] In a possible embodiment, along the extension direction of the tab, a distance between the first section and the bottom edge is a height Y of the sealing piece, where 3.5 mm≤Y≤12.5 mm.
[0042] In a possible embodiment, the second section and/or the third section includes the first subsection and the second subsection, where the first subsection is located between the first section and the second subsection, the second subsection includes at least one section, an endpoint connecting the first subsection and the first section is M_1, an endpoint connecting the second subsection and the side edge of the sealing piece is N_1, the connection endpoint M_1 and the connection endpoint N_1 form a line segment M_1N_1, and an included angle between the first section and the line segment M_1N_1 is greater than or equal to 90°.
[0043] In a possible embodiment, the second subsection includes at least two sections, and the included angle between adjacent sections is greater than or equal to 90°.
[0044] In a possible embodiment, the second subsection includes the n sections, and the n-th section is connected to the side edge of the sealing piece; after the die-cutting the top edge of the sealing piece, a distance from an intersection between the n-th section to the side edge of the tab does not exceed 2 mm; and after the step of hot pressing and sealing the packaging bag, the distances from the intersections between the (n−1)th section close to the first subsection and the n-th section to the side edge of the tab do not exceed 2.5 mm.
[0045] In this embodiment of the manufacturing method, the beneficial effects of various parameters of the sealing piece after hot pressing and sealing are roughly the same as those in the previous embodiments and will not be repeated herein. Before hot pressing and sealing, a cutting size parameter of part of the sealing piece is greater than that of part of the sealing piece after hot pressing and sealing, to reserve a certain deformation margin and tolerance range for the subsequent hot melting deformation of the sealing piece during the cutting stage, allowing the deformation amount of the sealing piece to be within a controllable range, thereby avoiding problems such as the sealing piece extending outward or warping after hot sealing.
[0046] According to the electrochemical apparatus, the electronic device, and the manufacturing method of this application, the height difference between the second section and the first section and the height difference between the third section and the first section are formed, such that the second section and the third section do not exceed the first section. The height difference between the second section and the first section and the height difference between the third section and the first section can reduce the impact caused by the outward-extending deformation of the sealing piece, helping reduce the exposed length of the sealing piece. After the tab is bent, because the sealing piece does not have a warping problem, the sealing piece has a reduced exposed length, and does not protrude relative to the battery cell in the thickness direction of the battery cell, thereby reducing the overall length and thickness of the electrochemical apparatus, and increasing the energy density of the electrochemical apparatus.

Description

BRIEF DESCRIPTION

[0047] To describe the technical solutions of the embodiments of this application more clearly, the following briefly describes the accompanying drawings in the embodiments. It is appreciated that the accompanying drawings below show merely some embodiments of this application and thus should not be considered as limitations on the scope. Persons of ordinary skill in the art may still derive other related drawings from the accompanying drawings without creative efforts.

[0048] FIG. **1** is a schematic structural diagram of an electrochemical apparatus according to an embodiment of this application.

[0049] FIG. **2** is a front view of the electrochemical apparatus shown in FIG. **1**.

[0050] FIG. **3** is a schematic structural diagram of a tab and a sealing piece in the electrochemical apparatus shown in FIG. **1**.

[0051] FIG. **4** is a front view of a structure shown in FIG. **3**.

[0052] FIG. **5** is a schematic structural diagram of a battery cell and a sealing piece in the electrochemical apparatus shown in FIG. **1**.

[0053] FIG. **6** is a front view of a structure shown in FIG. **5**.

[0054] FIG. **7** is a schematic structural diagram of the electrochemical apparatus being assembled into an electronic apparatus as shown in FIG. **1**.

[0055] FIG. **8** is a front view of a structure shown in FIG. **7**.

[0056] FIG. **9** is a schematic structural diagram of the tab and the sealing piece of the electrochemical apparatus according to an embodiment.

[0057] FIG. **10** is a schematic structural diagram of the tab and the sealing piece of the electrochemical apparatus according to an embodiment.

[0058] FIG. **11** is a schematic structural diagram of the tab and the sealing piece of the electrochemical apparatus according to an embodiment.

[0059] FIG. **12** is a schematic structural diagram of the tab and the sealing piece of the electrochemical apparatus according to an embodiment.

[0060] FIG. **13** is a schematic structural diagram of the tab and the sealing piece of the electrochemical apparatus according to an embodiment.

[0061] FIG. **14** is a schematic structural diagram of a battery cell and a sealing piece in the prior art.

[0062] FIG. **15** is a schematic structural diagram of the sealing piece deforming (warping) after hot pressing and sealing of the electrochemical apparatus in the prior art.

[0063] FIG. **16** is a schematic structural diagram of assembly interference between an electrochemical apparatus and an electronic device in the prior art.

REFERENCE SIGNS OF MAIN COMPONENTS

[0064] Electrochemical apparatus **100** [0065] Battery cell **1** [0066] Electrode plate **11** [0067] Tab **12** [0068] Packaging bag **2** [0069] Sealing edge **21** [0070] Sealing piece **3** [0071] Glue overflow portion **31** [0072] Top edge **4** [0073] First section **41** [0074] Second section **42** [0075] Third section **43** [0076] Depression **44** [0077] First subsection **45** [0078] Second subsection **46** [0079] Groove **47** [0080] Bottom edge **5** [0081] Side edge **6** [0082] Circuit board **7** [0083] Electronic device **200** [0084] Housing **201** [0085] Accommodating cavity **202** [0086] Electronic component connection line **203**

[0087] This application will be further described with reference to the accompanying drawings in the following specific embodiments.

DESCRIPTION OF EMBODIMENTS

[0088] The following clearly and completely describes the technical solutions in some embodiments of this application with reference to the accompanying drawings in some embodiments of this application. Apparently, the described embodiments are some but not all embodiments of this application.

[0089] It should be noted that when a component is referred to as being “fastened to” another component, it may be directly fastened to the another component, or there may be a component in between. When a component is deemed as being “connected to” another component, it may be directly connected to the another component, or there may be a component in between. When a component is deemed as being “provided on” another component, it may be directly provided on the another component, or there may be a component in between. The terms “vertical”, “horizontal”, “left”, “right”, and other similar expressions as used herein are for illustration only.

[0090] Unless otherwise defined, all technical and scientific terms used herein shall have the same meanings as commonly understood by persons skilled in the art to which this application belongs. The terms used herein in the specification of this application are merely intended to describe specific embodiments but not intended to limit this application. The term “and/or” used herein includes any and all combinations of one or more related listed items.

[0091] Some embodiments of this application are described in detail. Provided that there is no conflict, the following embodiments and features in an embodiment may be combined with each other.

[0092] Referring to FIGS. **1** to **4**, an embodiment provides an electrochemical apparatus **100**, including a battery cell **1**, a packaging bag **2**, and a sealing piece **3**. The battery cell **1** includes an electrode plate **11** and a tab **12** connected to the electrode plate **11**. The battery cell **1** is disposed in the packaging bag **2**, and the tab **12** protrudes from a sealing edge **21** of the packaging bag **2**. The sealing piece **3** is disposed on both side of the tab **12**, the packaging bag **2** covers part of the sealing piece **3**, the sealing piece **3** includes a top edge **4** and a bottom edge **5** opposite each other, the top edge **4** is located outside the packaging bag **2**, and the bottom edge **5** is located inside the packaging bag **2**. The top edge **4** includes a first section **41**, a second section **42**, and a third section **43**, the second section **42** and the third section **43** are respectively located on two sides of the first section **41**, and along a width direction of the tab **12**, a length of the first section **41** is A, a length of the bottom edge **5** is B, and a width of the tab **12** is C, meeting $B>A\geq C$. Along an extension direction of the tab **12**, a distance from any point on the second section **42** and the third section **43** to the bottom edge **5** is less than a distance between the first section **41** and the bottom edge **5**.

[0093] Thus, along the extension direction of the tab **12**, the second section **42** and the third section **43** do not exceed the first section **41**. The height difference between the second section **42**, the third section **43**, and the first section **41** can reduce the impact caused by the outward-extending deformation of the sealing piece **3**, reduce the exposed size and warping amount of the sealing piece **3**, and alleviate the problem of the top edge **4** of the sealing piece **3** deforming and warping due to excessive length of the top edge **4** during the assembly of the electrochemical apparatus **100**, otherwise such problem leads to interference between the electrochemical apparatus **100** and the battery compartment. In addition, reducing the warping amount in the thickness direction of the electrochemical apparatus **100** can reduce the loss of energy density of the electrochemical apparatus **100**. After the tab **12** is bent, because there is no warping problem with the sealing piece **3**, and the size of the sealing piece **3** in the thickness direction of the electrochemical apparatus **100** is reduced, the overall thickness of the electrochemical apparatus **100** is reduced, increasing the energy density of the electrochemical apparatus **100**.

[0094] In an embodiment of this application, the sealing piece **3** shown in FIGS. **1** to **6** is formed by sandwiching two opposite sides of the tab **12** between two sealing films and then performing hot melting and sealing, such that the sealing piece **3** can surround the periphery of the tab **12**, which helps improve the sealing effect between the tab **12** and the packaging bag **2**. In other embodiments, the sealing piece **3** may alternatively be formed by placing a sealing film on the surface of the tab **12** and then performing hot melting and sealing, or by clamping the tab **12** with multiple sealing films to meet actual sealing requirements. This application is not limited thereto.

The size relationship between the above sections and the bottom edge is the size relationship after hot melting of the sealing piece.

[0095] In an embodiment of this application, the second section **42** is obliquely connected to an end of the first section **41** and a side edge **6** of the sealing piece **3**, and the third section **43** is obliquely connected to another end of the first section **41** and another side edge **6** of the sealing piece **3**, such that the second section **42** and the third section **43** form chamfered structures on two sides of the top edge **4** of the sealing piece **3**. The second section **42** and the third section **43** may be formed by cutting the top edge **4** of the sealing piece **3**, and the cutting process may be performed before the tab **12** is connected to the electrode plate **11**. In other words, before the packaging process, the second section **42** and the third section **43** have already been formed on the top edge **4** of the sealing piece **3**. Thus, when the exposed part of the sealing piece **3** deforms during the hot pressing process, the chamfered regions at the second section **42** and the third section **43** can absorb the deformation amount, alleviating the problem of the deforming edge extending beyond the first section **41** along the extension direction of the tab **12**, thereby avoiding an increase in the exposed length of the sealing piece **3**. It should be noted that the first section **41** is not necessarily straight. After hot pressing and sealing, the first section **41** may slightly protrude outward in the extension direction of the tab **12**. Under the pressure of the hot pressing equipment, the protrusion amount of the first section **41** is small.

[0096] A glue overflow portion **31** is further formed between the two sides of the tab **12** and the first section **41**, and is naturally formed during the hot pressing process of the sealing piece **3**. In actual production, the size and proportion of the glue overflow portion **31** are much less than those shown in the drawings. After being bent, the tab **12** shields the glue overflow portion **31**, or the glue overflow portion **31** bends along with the tab **12**. The impact of the glue overflow portion **31** on the size of the electrochemical apparatus **100** may be ignored. The accompanying drawings of this application are to show the structure of the glue overflow portion **31**, and size and proportion of the glue overflow portion **31** in the drawings are enlarged.

[0097] Still referring to FIGS. **5** and **6**, the battery cell **1** further includes a separator, and the electrode plate **11** further includes a positive electrode plate and a negative electrode plate. The separator is placed between the positive electrode plate and the negative electrode plate to avoid short circuit problems caused by contact between the positive and negative electrode plates. The tab **12** is provided in two. One tab **12** is connected to the positive electrode plate, and the other tab **12** is connected to the negative electrode plate. Each tab **12** is provided with a corresponding sealing piece **3**, and each sealing piece **3** is formed by sandwiching the tab **12** between at least two sealing films and then performing hot melting. In an embodiment of this application, the positive electrode plate, the separator, and the negative electrode plate are wound or stacked to form the battery cell **1**, and the two tabs **12** are two located at the same end of the battery cell **1**. It may be understood that in other embodiments, the two tabs **12** may alternatively be located at different ends of the battery cell **1**, and the number of tabs **12** may be more than two. This application is not limited thereto. The positive electrode plate, the separator, and the negative electrode plate may alternatively form the battery cell **1** by stacking multiple layers. This application does not limit the arrangement form of the electrode plate **11** of the battery cell **1**.

[0098] As shown in FIG. **4**, in a possible embodiment of this application, $A<B$, and the difference between the length B of the bottom edge **5** of the sealing piece **3** and the length A of the first section **41** is $B-A$, where $0<B-A\leq 8$ mm, such that the length A of the first section **41** is less than the length B of the bottom edge **5**, and the second section **42** and the third section **43** may be formed on two sides of the first section **41**. If the width of the bottom edge **5** is constant, when the value of $B-A$ is too large, the difference between the length of the first section **41** and the width of the tab **12** is too small, which is likely to cause cracking at the joint between the sealing piece **3** and the tab during assembly, form a leakage channel, and increase the electrolyte leakage rate of the electrochemical apparatus **100**. Additionally, an increase in the value of $B-A$ also increases the risk

of the second section **42** or the third section **43** of the sealing piece **3** entering the sealing part of the packaging bag **2** of the electrochemical apparatus **100**, thereby increasing the electrolyte leakage rate of the electrochemical apparatus **100**. Further, 1.8 mm≤B−A≤8 mm, such that the two ends of the first section **41** can form distinct second section **42** and third section **43**, reducing the exposed size and warping amount of the sealing piece **3**, alleviating interference problems during assembly, and reducing the loss of energy density of the electrochemical apparatus, while not causing a high electrolyte leakage rate of the electrochemical apparatus. Further, in an embodiment, the difference B−A between the length B of the bottom edge **5** of the sealing piece **3** and the length A of the first section **41** is preferably 3 mm≤B−A≤7 mm, which helps further reduce the exposed size of the sealing piece **3** and ensures the sealing performance of the electrochemical apparatus **100**. It should be noted that the parameter range of the difference B−A between the length B of the bottom edge **5** and the length A of the first section **41** is the parameter range after hot sealing of the sealing piece **3**, to ensure that the top edge **4** of the sealing piece **3** has no part extending outward and deforming, achieving the objective of reducing the exposed size of the sealing piece **3**. When the length A is measured, due to the influence of hot melting, there is a protrusion on the top edge **4** in the extension direction of the tab **12**. Two endpoints of the first section **41** are selected at the turning positions at which the distances from the top edges **4** on two sides of the tab **12** to the bottom edge **5** decrease, and the distance between the two endpoints is the length A of the first section **41**. When the length B is measured, the distance between the two side edges **6** of the sealing piece **3** is measured. If the turning positions on the two sides of the tab cannot be determined, the length A of the first section **41** may be tested after downward movement by 0.04 mm from the highest point of the bulge on the top edge **4**.

[0099] In an embodiment of this application, along the extension direction of the tab **12**, a distance between two ends of the second section **42** or the third section **43** is a height H of the second section **42** or the third section **43**, where 0.05 mm≤H≤0.5 mm. In an embodiment of this application, the distance H between the two ends of the second section **42** is the vertical distance between the left endpoint of the top edge **4** and the left endpoint of the first section **41** along the extension direction of the tab **12**; and the distance H between the two ends of the third section **43** is the vertical distance between the right endpoint of the top edge **4** and the right endpoint of the first section **41** along the extension direction of the tab **12**. The measurement method for the first section **41** is described in the above embodiments and is not elaborated herein. The parameter range of H is also the height parameter range of the second section **42** and/or the third section **43** of the sealing piece **3** after hot melting. To be specific, the height difference between the second section **42** and the first section **41** and the height difference between the third section **43** and the first section **41** after hot sealing are in the range of 0.05 mm-0.5 mm, ensuring that the second section **42** and the third section **43** do not exceed the first section **41**, which can reduce the impact caused by the outward-extending deformation of the sealing piece **3**, thereby achieving the objective of reducing the exposed size of the sealing piece **3**, and alleviating interference problems during subsequent assembly. In addition, the height difference between the second section **42** and the first section **41** and the height difference between the third section **43** and the first section **41** allow part of the sealing piece **3** within this range to bend together with the tab **12**, thereby reducing the size of the electrochemical apparatus **100** in the length direction and increasing the energy density of the electrochemical apparatus **100**. When the value of H exceeds the preset range, a larger value of H indicates a greater risk of the second section **42** and the third section **43** of the sealing piece **3** entering the sealing part of the packaging bag **2**, thus reducing the sealing performance of the electrochemical apparatus **100** and increasing the electrolyte leakage rate. In an embodiment of this application, the heights of the second section **42** and the third section **43** are the same and are symmetrically arranged on two sides of the first section **41**. This symmetrical arrangement helps maintain the uniformity of the packaging effect and reduce the preparation process. In other embodiments, the heights of the second section **42** and the third section **43** may alternatively be different to meet actual design requirements.

[0100] Further, in an embodiment, the height H of the second section **42** or the third section **43** is preferably 0.15 mm≤H≤0.3 mm. This parameter range is also the height parameter range of the second section **42** and/or the third section **43** of the sealing piece **3** after hot melting, so as to further reduce the exposed size of the sealing piece **3**. This can improve the energy density performance of the electrochemical apparatus **100**, alleviate the leakage problem, and meet subsequent assembly requirements.

[0101] In a direction from a side edge **6** of the sealing piece **3** to another side edge **6**, that is, along the width direction P of the sealing piece **3**, a distance from one end of the second section **42** or the third section **43** connected to the first section **41** to the tab **12** is W, meaning that a shoulder width of the top edge of the sealing piece **3** is W, where 0.3 mm≤W≤1.1 mm. This parameter range is also the parameter range of the shoulder width of the top edge **4** of the sealing piece **3** after hot melting of the sealing piece **3**, which helps maintain the covering effect of the sealing piece **3** on the tab **12** and ensures the sealing performance between the tab **12** and the packaging bag **2**.

[0102] Further, in a possible embodiment of this application, along the width direction P of the sealing piece **3**, the distance between the side edge **6** of the sealing piece **3** and the corresponding side edge **6** of the tab **12** is X, meaning that the shoulder width of the bottom edge of the sealing piece **3** is X, where 0.3 mm≤X≤4.5 mm, such that an end of the bottom edge **5** of the sealing piece **3** exceeds an end of the first section **41**, increasing the contact area of the sealing piece **3** in the packaging bag **2**, which helps ensure the sealing effect between the tab **12** and the packaging bag **2**.

[0103] Along the extension direction of the tab **12**, the distance between the first section **41** and the bottom edge **5** is the height Y of the sealing piece **3**, where 3.5 mm≤Y≤12.5 mm. A suitable value may be chosen based on the size of the tab **12** to maintain the sealing effect between the tab **12** and the packaging bag **2** while reasonably planning the height dimensions of the sealing piece **3**, alleviating the problem of the increase in the overall size of the electrochemical apparatus **100** caused by excessive exposure of the sealing piece **3**.

[0104] Stilling referring to FIG. **2**, in an embodiment of this application, along the extension direction of the tab **12**, the distance Z between an end of the second section **42** away from the first section **41** and the sealing edge **21** of the packaging bag **2** and the distance Z between an end of the third section **43** away from the first section **41** and the sealing edge **21** of the packaging bag **2** are both greater than or equal to 0.1 mm, reducing the risk of poor sealing and electrolyte leakage caused by embedding the second section or the third section into the sealing region of the packaging bag **2**. Along the extension direction of the tab **12**, the exposed length of the sealing edge **21** of the sealing piece **3** is D, where Z<D, such that the top edge **4** of the sealing piece **3** can form the second section **42** and the third section **43**. Further, the packaging bag **2** of this application is approximately rectangular. Along the extension direction of the tab **12**, the vertical distances from any points on the sealing edge **21** to the first section **41** of the top edge **4** of the sealing piece **3** are the same. Thus, the shape of the sealing edge **21** is regular and stable, which can balance the sealing effect of various parts of the sealing region of the packaging bag **2**, helping improve the sealing effect, reduce stress concentration problems, and reduce the risk of weak sealing regions and leakage. It should be noted that the vertical distances from any points on the sealing edge **21** to the first section **41** of the top edge **4** of the sealing piece **3** are the same, which does not mean the distances are absolutely identical. When comparing the vertical distances from any two points on the sealing edge **21** to the first section **41**, vertical distances with an error of within +3% may be considered the same.

[0105] Table 1 and Table 2 show the dimensional parameters of the sealing piece before and after hot pressing in each example, including: the length A of the first section **41**, the length B of the bottom edge **5**, the difference B−A between the lengths of the first section **41** and the bottom edge **5**, the height H of the second section **42** and the third section **43**, the shoulder width X of the bottom edge of the sealing piece **3**, the shoulder width W of the top edge of the sealing piece **3**, and the length D of the sealing piece **3** exposed from the sealing edge **21**.

TABLE-US-00001 TABLE 1 Dimensional parameters of sealing pieces in various examples before hot pressing Before hot pressing (in mm) H.sub.1 X W.sub.1 B.sub.1 − A.sub.1 0.15 ≤ 0.3 < 0.2 ≤ 0.2 < B − H.sub.1 ≤ X ≤ W ≤ A.sub.1 B.sub.1 A ≤ 8.3 0.7 4.5 1.3 D Example 1 2.7 11 8.3 0.5 4.3 0.45 1 Example 2 10.5 11 0.5 0.5 0.4 0.45 1 Example 3 6.3 11 4.7 0.7 2.5 0.45 1 Example 4 7.7 11 3.3 0.5 1.8 0.45 1 Example 5 7.7 11 3.3 0.15 1.8 0.45 1 Example 6 7.7 11 3.3 0.6 1.8 0.45 1 Example 7 7.7 11 3.3 0.3 1.8 0.45 1 Example 8 7.7 11 3.3 0.3 1.8 0.45 1 Example 9 7.7 11 3.3 0.5 2.6 1.25 1 Example 10 3.7 11 7.3 0.5 3.8 0.45 1 Example 11 8.9 11 2.1 0.5 1.2 0.45 1 Example 12 9.7 11 1.3 0.5 0.8 0.45 1 Example 13 8.3 11 2.7 0.5 1.5 0.45 1 Example 14 6.5 11 4.5 0.5 2.4 0.45 1 Example 15 4.7 11 6.3 0.5 3.3 0.45 1 Example 16 3.3 11 7.7 0.5 4 0.45 1 Example 17 8.3 11 2.7 0.5 1.8 0.75 1 Example 18 8.9 11 2.1 0.5 1.8 1.05 1 Example 19 7.7 11 3.3 0.5 1.8 0.45 1 Example 20 2.5 11 8.5 0.5 4.4 0.45 1 Comparative 7.7 11 3.3 0.8 1.8 0.45 1 example 1 Comparative 7.7 11 3.3 0.24 1.8 0.45 1 example 2 Comparative 11 11 0 0 2.5 2.5 1 example 3

TABLE-US-00002 TABLE 2 Dimensional parameters and technical effects of sealing pieces in various examples after hot pressing After hot pressing (in mm) B − A H X W Energy 0 < B − 0.05 ≤ 0.3 < 0.3 ≤ density Electrolyte A B A ≤ 8 H ≤ 0.5 X ≤ 4.5 W ≤ 1.1 D (Wh/L) leakage rate Example 1 3 11 8 0.3 4.3 0.3 1 708.52 0.000265% Example 2 10.8 11 0.2 0.3 0.4 0.3 1 706.27 0.000124% Example 3 6.6 11 4.4 0.5 2.5 0.3 1 709.09 0.000274% Example 4 8 11 3 0.3 1.8 0.3 1 707.66 0.000200% Example 5 8 11 3 0.05 1.8 0.3 1 705.70 0.000125% Example 6 8 11 3 0.4 1.8 0.3 1 708.50 0.000220% Example 7 8 11 3 0.15 1.8 0.3 1 706.83 0.000155% Example 8 8 11 3 0.1 1.8 0.3 1 706.41 0.000148% Example 9 8 11 3 0.3 2.6 1.1 1 706.55 0.000185% Example 10 4 11 7 0.3 3.8 0.3 1 708.10 0.000247% Example 11 9.2 11 1.8 0.3 1.2 0.3 1 706.97 0.000145% Example 12 9.4 11 1.6 0.3 1 0.2 9.4 706.69 0.000135% Example 13 8.6 11 2.4 0.3 1.5 0.3 1 707.11 0.000168% Example 14 6.8 11 4.2 0.3 2.4 0.3 1 707.82 0.000225% Example 15 5 11 6 0.3 3.3 0.3 1 707.96 0.000235% Example 16 3.6 11 7.4 0.3 4 0.3 1 708.24 0.000250% Example 17 8.6 11 2.4 0.3 1.8 0.6 1 707.25 0.000192% Example 18 9.2 11 1.8 0.3 1.8 0.9 1 707.39 0.000188% Example 19 8 11 2.8 0.3 1.8 0.3 1 707.53 0.000185% Example 20 2.8 11 8.2 0.3 4.4 0.3 1 709.09 0.000305% Comparative 8 11 3 0.6 1.8 0.3 1 709.65 0.000505% example 1 Comparative 8 11 3 0.04 1.8 0.3 1.05 705.28 0.000120% example 2 Comparative 8 11 3 −0.4 2.5 2.5 1.4 704.43 0.000115% example 3

[0106] From the comparison of the example data in Table 1 and Table 2, it can be seen that although the sealing piece **3** undergoes slight deformation after hot pressing, for example, the length A of the first section **41** increases, the heights H of the second section **42** and the third section **43** decrease, and the shoulder width W of the top edge of the sealing piece **3** increases, all parameters of the sealing piece **3** after hot pressing can still meet the foregoing conditions, and the length D of the sealing piece **3** protruding from the sealing side edge remains unchanged, being 1 mm before and after hot pressing. In the comparative examples, the sealing piece is rectangular, with the top edge and bottom edge having the same length, and the shoulder width of the top edge is also the same as the shoulder width of the bottom edge. The second section and the third section are not formed on two sides of the top edge. After hot pressing, the heights H of the second section **42** and the third section **43** protrude 0.4 mm along the extension direction of the tab, causing the length D of the sealing piece protruding from the sealing edge to significantly increase from 1 mm to 1.4 mm. This increases the overall length of the electrochemical apparatus **100**, resulting in warping in the thickness direction, affecting energy density, and causing interference problems between the electrochemical apparatus **100** and the battery compartment during assembly.

[0107] Through data comparison between the examples and comparative examples, along the extension direction of the tab **12**, a distance between two ends of the second section **42** or the third section **43** is a height H of the second section **42** or the third section **43**, where 0.05 mm≤H≤0.5 mm. By ensuring that the second section **42** and the third section **43** do not exceed the first section **41**, the impact caused by the outward-extending deformation of the sealing piece **3** can be reduced, thereby achieving the objective of reducing the exposed size of the sealing piece **3** and alleviating interference problems during subsequent assembly processes. In addition, the height difference between the second section **42** and the first section **41** and the height difference between the third section **43** and the first section **41** allow part of the sealing piece **3** within this range to bend together with the tab **12**, thereby reducing the size of the electrochemical apparatus **100** in the length direction and increasing the energy density of the electrochemical apparatus **100**. The data from Example 20, Comparative example 1, and Comparative example 2 demonstrate that when the value of H exceeds the preset range, a larger value of H indicates a greater risk of the second section **42** and the third section **43** of the sealing piece **3** entering the sealing part of the electrochemical apparatus **100**, which reduces the sealing performance of the electrochemical apparatus **100** and increases the probability of leakage. Further, the height H of the second section **42** or the third section **43** is preferably 0.15 mm≤H≤0.3 mm. This parameter range is also the height parameter range of the second section **42** and/or the third section **43** of the sealing piece **3** after hot melting. This can further reduce the exposed size of the sealing piece **3**, improve the energy density performance, alleviate leakage problems of the electrochemical apparatus **100**, and meet subsequent assembly requirements. A<B, the difference between the length B of the bottom edge **5** of the sealing piece **3** and the length A of the first section **41** is B−A, and 0<B−A≤8 mm, such that the length A of the first section **41** is less than the length B of the bottom edge **5**, and the second section **42** and the third section **43** can be formed on two sides of the first section **41**. When the value of B−A is too large, that is, the difference between the length of the first section **41** and the width of the tab **12** is too small, it is easy to cause the tab adhesive to crack during assembly, forming a leakage channel and increasing the electrolyte leakage rate of the electrochemical apparatus. Additionally, an increase in the value of B−A also increases the risk of the second section **42** or the third section **43** of the sealing piece entering the sealing part of the packaging bag **2** of the electrochemical apparatus **100**, thereby increasing the electrolyte leakage rate of the electrochemical apparatus **100**. Further, 1.8 mm≤B−A≤8 mm, such that the two ends of the first section **41** can form distinct second section **42** and third section **43**, reducing the exposed size and warping amount of the sealing piece **3**, alleviating the interference problems during assembly, and reducing the loss of energy density of the electrochemical apparatus **100**, while not causing the electrolyte leakage rate of the electrochemical apparatus **100** to be too high. Further, in an embodiment, the difference B−A between the length B of the bottom edge **5** of the sealing piece **3** and the length A of the first section **41** is preferably 3 mm≤B−A≤7 mm, which helps further reduce the exposed size of the sealing piece **3** and ensures the sealing performance of the electrochemical apparatus **100**. In summary, it is sufficient to prove that the sealing piece structure in some embodiments of this application can reduce the impact caused by the outward-extending deformation of the top edge, reduce the exposed size of the sealing size **3**, and alleviate the problem of the top edge **4** of the sealing piece **3** deforming and warping due to excessive length during the assembly of the electrochemical apparatus **100**, otherwise such problem leads to interference between the electrochemical apparatus **100** and the battery compartment. Additionally, the sealing piece structure can also reduce the high electrolyte leakage rate of the electrochemical apparatus **100**, otherwise such high rate affects the production yield of the electrochemical apparatus **100**.

[0108] Referring to FIGS. **7** and **8**, in a possible embodiment of this application, the electrochemical apparatus **100** further includes a circuit board **7**, where the circuit board **7** is located on a side of the sealing piece **3** and covers part of the sealing piece **3** exposed by the packaging bag **2**, and an end of the tab **12** protruding from the packaging bag **2** is bent toward the circuit board **7** and connected to the circuit board **7**. Because the arrangement of the second section **42** and the third section **43** can reduce the impact caused by the outward-extending deformation of the top edge **4** of the sealing piece **3**, along the extension direction of the tab **12**, the second section **42** and the third section **43** do not exceed the first section **41**. When the circuit board **7** covers the first section **41**, it may cover the exposed part of the sealing piece **3**, and the top edge **4** of the sealing piece **3** does not exceed the edge of the circuit board **7**. Therefore, the overall size of the electrochemical apparatus **100** is reduced, helping to increase the energy density of the electrochemical apparatus **100**. During assembly of the electrochemical apparatus **100** into the electronic device **200**, interference problems are also less likely to occur, reducing space occupied by the electrochemical apparatus **100** in the electronic device **200**.

[0109] Referring to FIGS. **9** and **10**, in a possible embodiment of this application, the second section **42** is an arc section connected between an end of the first section **41** and a side edge **6** of the sealing piece **3**, and the second section **42** is recessed toward a side edge **6** of the tab **12** or the second section **42** is bent toward an outer side of the sealing piece **3**. A rounded corner structure is formed at the end of the top edge **4** of the sealing piece **3**. The rounded corner structure can reduce the sharp parts presented at the end of the top edge **4** of the sealing piece **3**, reducing the warping of the sealing piece. The second section **42** does not exceed the first section **41** along the extension direction of the tab **12**, and the shoulder width W of the top edge of the sealing piece **3** and the height H of the second section **42** meet the parameter requirements of the aforementioned embodiments, achieving the objective of reducing the exposed size of the sealing piece **3**.

[0110] The third section **43** is an arc section connected between another end of the first section **41** and another side edge **6** of the sealing piece **3**, and the third section **43** is recessed toward the side edge **6** of the tab **12** or the third section **43** is bent toward the outer side of the sealing piece **3**. The third section **43** does not exceed the first section **41** along the extension direction of the tab **12**, and the shoulder width W of the top edge of the sealing piece **3** and the height H of the third section **43** meet the parameter requirements of the aforementioned embodiments, achieving the objective of reducing the exposed size of the sealing piece **3**. In an embodiment of this application, the second section **42** and the third section **43** are symmetrically arranged at two ends of the first section **41**, either as rounded corner structures bent outward or as rounded corner structures recessed inward. In other embodiments, the second section **42** and the third section **43** may alternatively be one as a rounded corner structure bent outward and the other as a rounded corner structure recessed inward, to meet actual design requirements.

[0111] Referring to FIG. **11**, the second section **42** and/or the third section **43** is provided with a depression **44**, such that the second section **42** and/or the third section **43** includes a first subsection **45** and a second subsection **46**. The first subsection **45** is located between the first section **41** and the second subsection **46**, the second subsection **46** includes at least one section, an endpoint connecting the first subsection **45** and the first section **41** is M, an endpoint connecting the second subsection **46** and the side edge of the sealing piece **3** is N, the connection endpoint M and the connection endpoint N form a line segment MN, and an included angle between the first section **41** and the line segment MN is greater than or equal to 90°.

[0112] In an embodiment of this application, the first subsection **45** and the second subsection **46** are respectively two adjacent sides of the depression **44**, and the included angle between the first subsection **45** and the second subsection **46** is less than 180°, such that the second section **42** and the third section **43** have a stepped surface structure recessed toward the side edge **6** of the tab **12**, which can further reduce the impact caused by the outward-extending deformation of the sealing piece **3** and reduce the exposed size of the sealing piece **3**. In other embodiments, as shown in FIG. **13**, the second subsection **46** includes at least two sections, and the included angle between adjacent sections is greater than or equal to 90°. This multi-section structure can cause the deformation amount of the sealing piece **3** extending outward to be mainly within the plane range of the electrochemical apparatus **100**, reducing the warping amount of the sealing piece **3** in the thickness direction of the electrochemical apparatus **100**, thus helping reduce the loss of energy density. In a possible embodiment of this application, the second subsection **46** includes n sections, the n-th section is connected to the side edge of the sealing piece **3**, the distances from the intersections between the (n−1)th section close to the first subsection **45** and the n-th section to the side edge of the tab **12** do not exceed 2.5 mm. This multi-section arrangement allows the top edge **4** of the sealing piece **3** to fit better, improving the sealing performance of the electrochemical apparatus.

[0113] Further, referring to FIG. **12**, a groove **47** is further provided between the first subsection **45** and the second subsection **46**, the structure of the groove **47** may be arc-shaped, rectangular, polygonal, or the like, for absorbing the deformation of the sealing piece **3**, and under the action of the groove **47**, when the sealing piece **3** is hot pressed to deform, the first subsection **45** and the second subsection **46** may approach each other, reducing the risk caused by the outward-extending deformation of the top edge **4**.

[0114] Still referring to FIGS. **7** and **8**, an embodiment of this application also provides an electronic device **200**. The electronic device **200** includes a housing **201**, an electronic component connection line **203**, and the electrochemical apparatus **100** as described in the above embodiments, where the electronic component connection line **203** and the electrochemical apparatus **100** are disposed in the housing **201**, and the electronic component connection line **203** is electrically connected to the battery cell **1** in the electrochemical apparatus **100** via a circuit board **7**. Specifically, the housing **201** is provided with an accommodating cavity **202**, the electrochemical apparatus **100** is mounted in the accommodating cavity **202**, and the top edge **4** of the sealing piece **3** of the electrochemical apparatus **100** does not exceed the accommodating cavity **202**. This arrangement of the second section **42** and the third section **43** alleviates the problem caused by the outward-extending warping of the top edge **4** of the sealing piece **3** after hot pressing, thereby preventing interference problems when the electrochemical apparatus **100** is placed in the accommodating cavity **202**, which can reduce the size of the accommodating cavity **202** to some extent, contributing to the miniaturization of the electronic device **200**.

[0115] According to the electrochemical apparatus **100** and the electronic device **200** of this application, the height difference between the second section **42** and the first section **41** and the height difference between the third section **43** and the first section **41** are formed, such that after hot pressing and sealing, the second section **42** and the third section **43** do not exceed the first section **41**. The height difference between the second section **42** and the first section **41** and the height difference between the third section **43** and the first section **41** can offset the deformation amount of the sealing piece **3** extending outward, thereby alleviating the problem of two sides of the top end of the sealing piece **3** warping, and preventing an increase in the exposed length of the sealing piece **3**. After the tab **12** is bent, because the sealing piece **3** does not have a warping problem, the sealing piece has a reduced exposed length, and does not protrude relative to the battery cell in the thickness direction of the battery cell, thereby reducing the overall length and thickness of the electrochemical apparatus **100**, and increasing the energy density of the electrochemical apparatus **100**.

[0116] An embodiment of this application provides a manufacturing method for electrochemical apparatus, used for manufacturing the electrochemical apparatus **100** in the above embodiments. The manufacturing method for electrochemical apparatus includes the following steps: [0117] die-cutting the top edge **4** of the sealing piece **3**, such that the top edge **4** of the sealing piece **3** forms the first section **41**, the second section **42**, and the third section **43**, where the second section **42** and the third section **43** are respectively located on two sides of the first section **41**, along the width direction of the tab **12**, the length of the first section **41** is A, the length of the bottom edge **5** is B, and the width of the tab **12** is C, satisfying $B>A\geq C$; and along the extension direction of the tab **12**, the distance from any point on the second section **42** and the third section **43** to the bottom edge **5** is less than a distance between the first section **41** and the bottom edge **5**; [0118] preparing the tab **12**, and adhering the sealing piece **3** to the tab **12**; [0119] welding the tab **12** to the electrode plate **11**, winding or stacking the electrode plate to form the battery cell **1**, and placing the battery cell **1** into the packaging bag **2**, where the tab **12** protrudes from the sealing edge **21** of the packaging bag **2**, the packaging bag **2** covers part of the sealing piece **3**, the top edge **4** of the sealing piece **3** is located outside the packaging bag, and the bottom edge **5** is located inside the packaging bag; and

[0120] hot pressing and sealing the packaging bag **2**, where the sealing piece **3** is sealingly connected between the packaging bag **2** and the tab **12**.
[0121] In an embodiment of this application, the material of the sealing piece **3** includes polypropylene. The thickness of the sealing piece **3** is 30 μm-80 μm, and the melting temperature is 123±5° C.-158±10° C. The sealing piece **3** within this thickness range can provide effective sealing without excessively increasing the thickness of the sealing region of the packaging bag **2**, leaving enough space for the subsequent installation of the circuit board.
[0122] In the hot pressing and sealing step, the hot pressing pressure is 120±50 Kgf, and the hot pressing temperature is 190±5° C. The extension amount of the sealing piece **3** may be slightly adjusted based on the hot pressing parameters. Higher temperature, greater pressure, or longer hot pressing time causes a greater extension amount. Limiting the hot pressing parameters in this application can minimize the extension amount of the sealing piece while the sealing piece is sealingly connected between the tab and the packaging bag, and also minimize the impact of the extension amount on the exposed length of the sealing piece.
[0123] In an embodiment of this application, after hot pressing and sealing, the difference between the length B of the bottom edge **5** of the sealing piece **3** and the length A of the first section **41** is $0<B-A\leq 8$ mm, such that the length A of the first section **41** is less than the length B of the bottom edge **5**, and the second section **42** and the third section **43** can be formed on two sides of the first section **41**. The height H of the second section **42** and the third section **43** meets 0.05 mm≤H≤0.5 mm, ensuring that the second section **42** and the third section **43** do not exceed the first section **41**, which can reduce the impact caused by the outward-extending deformation of the sealing piece, thereby achieving the objective of reducing the exposed size of the sealing piece, and alleviating interference problems during subsequent assembly. The shoulder width W of the top edge of the sealing piece **3** meets 0.3 mm≤W≤1.1 mm, helping maintain the covering effect of the sealing piece **3** on the tab **12**, thus ensuring the sealing performance between the tab **12** and the packaging bag **2**. The shoulder width X of the bottom edge **5** of the sealing piece **3** meets 0.3 mm≤X≤4.5 mm, such that an end of the bottom edge **5** of the sealing piece **3** exceeds an end of the first section **41**, increasing the contact area of the sealing piece **3** in the packaging bag **2**, which helps ensure the sealing effect between the tab **12** and the packaging bag **2**. The height Y of the sealing piece **3** meets 3.5 mm≤Y≤12.5 mm, to maintain the sealing effect between the tab **12** and the packaging bag **2** while reasonably planning the height dimensions of the sealing piece **3**, alleviating the problem of the increase in the overall size of the electrochemical apparatus **100** caused by excessive exposure of the sealing piece **3**.
[0124] Further, in an embodiment of this application, 1.8 mm≤B−A≤8 mm, such that the two ends of the first section **41** can form distinct second section **42** and third section **43**, reducing the exposed size of the sealing piece **3**, alleviating interference problems during assembly, and reducing the loss of energy density of the electrochemical apparatus. Further, in an embodiment, the difference B−A between the length B of the bottom edge **5** of the sealing piece **3** and the length A of the first section **41** is preferably 3 mm≤B−A≤7 mm, which helps further reduce the exposed size of the sealing piece **3**. It should be noted that the parameter range of the difference B−A between the length B of the bottom edge **5** and the length A of the first section **41** is the parameter range after hot sealing of the sealing piece **3**, to ensure that the top edge **4** of the sealing piece **3** has no part extending outward and deforming, achieving the objective of reducing the exposed size of the sealing piece **3**.
[0125] Further, in an embodiment, the height H of the second section **42** or the third section **43** is preferably 0.15 mm≤H≤0.3 mm. This parameter range is also the height parameter range of the second section **42** and/or the third section **43** of the sealing piece **3** after hot melting, so as to further reduce the exposed size of the sealing piece **3**, meeting subsequent assembly requirements.
[0126] In a possible embodiment of this application, the second section **42** and/or the third section **43** is provided with a depression **44**, such that the second section **42** and/or the third section **43**

includes a first subsection **45** and a second subsection **46**. The first subsection **45** is located between the first section **41** and the second subsection **46**, the second subsection **46** includes at least one section, an endpoint connecting the first subsection **45** and the first section **41** is M.sub.1, an endpoint connecting the second subsection **46** and the side edge of the sealing piece **3** is N.sub.1, the connection endpoint M.sub.1 and the connection endpoint N.sub.1 form a line segment M.sub.1N.sub.1, and an included angle between the first section **41** and the line segment M.sub.1N.sub.1 is greater than or equal to 90°.

[0127] In a possible embodiment of this application, the second subsection **46** includes at least two sections, and the included angle between adjacent sections is greater than or equal to 90°. This multi-section structure can cause the deformation amount of the sealing piece **3** extending outward to be mainly within the plane range of the electrochemical apparatus **100**, reducing the warping amount of the sealing piece **3** in the thickness direction of the electrochemical apparatus **100**, thus helping to reduce the loss of energy density.

[0128] Further, the second subsection **46** includes the n sections, and the n-th section is connected to the side edge of the sealing piece **3**. Before the hot pressing and sealing process, when the second subsection **46** is cut into n sections, the distance from the intersection between the n-th section to the side edge of the tab **12** does not exceed 2 mm, so as to provide a certain deformation space for the sealing piece. After the hot pressing and sealing process, the distances from the intersections between the (n−1)th section close to the first subsection **45** and the n-th section to the side edge of the tab **12** do not exceed 2.5 mm. This multi-section arrangement allows the top edge **4** of the sealing piece **3** to fit better, improving the sealing performance of the electrochemical apparatus.

[0129] Additionally, the second section **42** includes a first subsection **45** and a second subsection **46**, where the second subsection **46** includes n sections, and the n-th section of the second subsection **46** is formed by cutting the sealing piece before the step of bonding the sealing piece and the tab. The first subsection **45** and the (n−1)th section of the second subsection can increase the extension amount of the partial regions of the sealing piece **3** on two sides of the tab **12** along the extension direction of the tab **12** by adjusting the hot pressing pressure and hot pressing temperature. This facilitates tight fitting of the sealing piece **12**, reducing the electrolyte leakage rate.

[0130] In an embodiment of this application, after the cutting of the sealing piece and before the hot pressing and sealing step, the difference between the length B.sub.1 of the bottom edge of the sealing piece and the length A.sub.1 of the first section is 0.2<B.sub.1−A.sub.1≤8.3 mm. Further, the height H.sub.1 of the second section and the third section meets 0.15 mm≤H.sub.1≤0.7 mm, and the shoulder width W.sub.1 of the top edge of the sealing piece meets 0.2 mm≤W.sub.1≤1.3 mm. The shoulder width X of the bottom edge of the sealing piece meets 0.3 mm≤X≤4.5 mm. The height Y of the sealing piece meets 3.5 mm≤Y≤12.5 mm. Before hot pressing and sealing, a cutting size parameter of part of the sealing piece is greater than that of part of the sealing piece after hot pressing and sealing, to reserve a certain deformation margin and tolerance range for the subsequent hot melting deformation of the sealing piece during the cutting stage, allowing the deformation amount of the sealing piece to be within a controllable range, thereby avoiding problems such as the sealing piece extending outward or warping after hot sealing.

[0131] The foregoing embodiments are merely intended to describe the technical solutions of this application, but not intended to constitute any limitation. Although this application is described in detail with reference to the foregoing preferred embodiments, persons of ordinary skill in the art should understand that modifications or equivalent replacements can be made to the technical solutions of this application, without departing from the spirit and scope of the technical solutions of this application.

Claims

1. An electrochemical apparatus, comprising: a battery cell comprising an electrode plate and a tab connected to the electrode plate; a packaging bag, wherein the battery cell is disposed in the packaging bag, the tab extends away from the electrode plate and protrudes out from a sealing edge of the packaging bag in a first direction; and a sealing piece disposed on two sides of the tab, wherein the packaging bag covers a part of the sealing piece, the sealing piece comprises a top edge and a bottom edge opposite to each other in the first direction, the top edge is located outside the packaging bag, and the bottom edge is located inside the packaging bag; wherein the top edge comprises a first section, a second section, and a third section; the first section is located between the second section and the third section along a width direction of the tab; along the width direction of the tab, a length of the first section is A, a length of the bottom edge is B, and a width of the tab is C, and $B>A\geq C$, $1.8\ mm\leq B-A\leq 8\ mm$; wherein along the width direction of the tab, the first section completely overlaps the tab; along the first direction, a distance between any point on the second section and the bottom edge is less than a distance between any point on the first section and the bottom edge; along the first direction, a distance between any point on the third section and the bottom edge is less than the distance between any point on the first section and the bottom edge; and along the first direction, a distance between two ends of the second section or a distance between two ends of the third section is a height H of the second section or the third section, wherein $0.05\ mm\leq H\leq 0.5\ mm$.
2. The electrochemical apparatus according to claim 1, wherein $3\ mm\leq B-A\leq 7\ mm$.
3. The electrochemical apparatus according to claim 1, wherein $0.15\ mm\leq H\leq 0.3\ mm$.
4. The electrochemical apparatus according to claim 1, wherein along the width direction of the tab, a distance between the second section and the tab or a distance between the third section and the tab is W, wherein $0.3\ mm\leq W\leq 1.1\ mm$.
5. The electrochemical apparatus according to claim 1, wherein along the width direction of the tab, a distance between a side edge of the sealing piece and a corresponding side edge of the tab is X, wherein $0.3\ mm\leq X\leq 4.5\ mm$.
6. The electrochemical apparatus according to claim 5, wherein along the first direction, a distance between the first section and the bottom edge is a height Y of the sealing piece, wherein $3.5\ mm\leq Y\leq 12.5\ mm$.
7. The electrochemical apparatus according to claim 1, wherein the second section is obliquely connected between an end of the first section and a side edge of the sealing piece, and the third section is obliquely connected between another end of the first section and another side edge of the sealing piece.
8. The electrochemical apparatus according to claim 1, wherein the second section is an arc section connected between an end of the first section and a side edge of the sealing piece, and the second section is recessed toward a side edge of the tab or the second section is bent toward an outer side of the sealing piece.
9. The electrochemical apparatus according to claim 8, wherein the third section is an arc section connected between another end of the first section and another side edge of the sealing piece, and the third section is recessed toward the side edge of the tab or the third section protrudes toward the outer side of the sealing piece.
10. The electrochemical apparatus according to claim 1, wherein the second section and/or the third section comprises a first subsection and a second subsection, wherein the first subsection is located between the first section and the second subsection, the second subsection comprises at least one section, an endpoint connecting the first subsection and the first section is M, an endpoint connecting the second subsection and the side edge of the sealing piece is N, the connection endpoint M and the connection endpoint N form a line segment MN, and an included angle

between the first section and the line segment MN is greater than or equal to 90°.
11. The electrochemical apparatus according to claim 10, wherein the second subsection comprises at least two sections, and an included angle between adjacent sections in the at least two sections is greater than or equal to 90°.
12. The electrochemical apparatus according to claim 10, wherein the second subsection comprises n sections, an n-th section is connected to the side edge of the sealing piece, and distances from intersections between (n−1)th section close to the first subsection and the n-th section to the side edge of the tab do not exceed 2.5 mm.
13. The electrochemical apparatus according to claim 10, wherein a groove is provided between the first subsection and the second subsection to absorb deformation of the sealing piece.
14. The electrochemical apparatus according to claim 1, wherein along the first direction, a distance between an end of the second section away from the first section and the sealing edge of the packaging bag and a distance between an end of the third section away from the first section and the sealing edge of the packaging bag are both greater than or equal to 0.1 mm.
15. The electrochemical apparatus according to claim 1, wherein the electrochemical apparatus further comprises a circuit board, wherein the circuit board is located on a side of the sealing piece and covers a part of the sealing piece exposed by the packaging bag, and an end of the tab protruding out from the packaging bag is bent towards the circuit board and connected to the circuit board.
16. The electrochemical apparatus according to claim 1, wherein the packaging bag is rectangular, and along the first direction, vertical distances from any points on the sealing edge to the first section of the top edge are identical.
17. An electronic device, comprising a housing, an electronic component connection line, and the electrochemical apparatus according to claim 1, wherein the electronic component connection line and the electrochemical apparatus are disposed in the housing, and the electronic component connection line is electrically connected to the electrochemical apparatus; the housing comprises an accommodating cavity, the electrochemical apparatus is mounted in the accommodating cavity, and the top edge of the sealing piece of the electrochemical apparatus does not exceed the accommodating cavity.
18. A method for manufacturing the electrochemical apparatus of claim 1, method comprising: die-cutting the top edge of the sealing piece, such that the top edge of the sealing piece forms the first section, the second section, and the third section, wherein the second section and the third section are respectively located on two sides of the first section, and along the width direction of the tab, the length of the first section is A, the length of the bottom edge is B, and the width of the tab is C, meeting $B>A\geq C$; and along the first direction, the distance from any point on the second section and the third section to the bottom edge is less than a distance between the first section and the bottom edge; adhering the sealing piece to the tab; welding the tab to the electrode plate, and winding or stacking the electrode plate to form the battery cell; placing the battery cell into the packaging bag, wherein the tab protrudes from the sealing edge of the packaging bag, the packaging bag covers part of the sealing piece, the top edge of the sealing piece is located outside the packaging bag, and the bottom edge of the sealing piece is located inside the packaging bag; and hot pressing and sealing the packaging bag, wherein the sealing piece is sealingly connected between the packaging bag and the tab.
19. The method for manufacturing the electrochemical apparatus according to claim 18, wherein in the step of hot pressing and sealing the packaging bag, a hot pressing pressure is 120±50 Kgf, and a hot pressing temperature is 190±5° C.
20. The method for manufacturing the electrochemical apparatus according to claim 18, wherein after the die-cutting the top edge of the sealing piece, the distance between two ends of the second section or the third section is a height H.sub.1 of the second section or the third section, wherein 0.15 mm≤H.sub.1≤0.7 mm; and after the step of hot pressing and sealing the packaging bag, the

height of the second section or the third section is H, wherein 0.05 mm≤H≤0.5 mm; after the die-cutting the top edge of the sealing piece, a length of the bottom edge of the sealing piece is B.sub.1, and a length of the first section is A.sub.1, meeting 0.2<B.sub.1−A.sub.1≤8.3 mm; and after the step of hot pressing and sealing the packaging bag, 0<B−A≤8 mm.